

Chapter 7: Human Health and Diseases.

"Health is Wealth, But Cleanliness is the Key."

* Health :

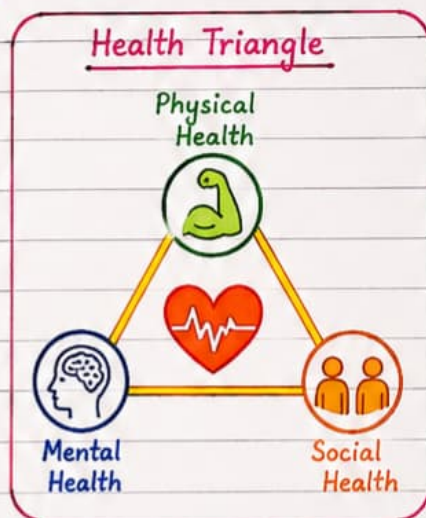
It is a state of complete **physical, mental and social well-being** and not merely the absence of disease or infirmity.

* Disease :

Any condition that disturbs the normal functioning of the body or mind is called disease.

* Types of Disease :

1. **Infectious Diseases** — Caused by pathogens.
2. **Non-infectious Diseases** — Not caused by pathogens.
3. **Deficiency Diseases** — Caused by lack of nutrients.



⇒ INFECTIOUS DISEASES ⇒

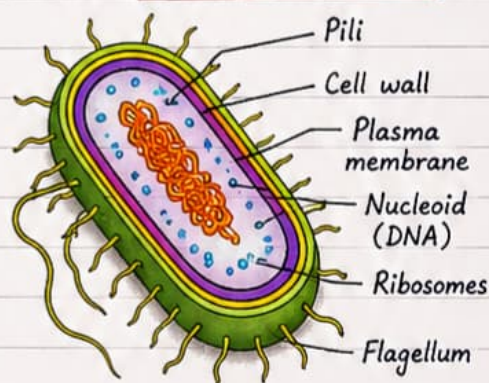
These diseases are caused by disease-causing organisms called **pathogens**. They can spread from one person to another.

Pathogens include :

- **Bacteria** (e.g., Mycobacterium, Salmonella)
- **Viruses** (e.g., HIV, Influenza virus)
- **Fungi** (e.g., Candida)
- **Protozoa** (e.g., Plasmodium)
- **Helminths** (e.g., Ascaris)

Different types of pathogens

Structure of a Bacterium (E.coli)



MODE OF SPREAD OF INFECTIOUS DISEASES

1. **Air borne** — Through air (e.g., Common cold, TB)
2. **Water borne** — Through contaminated water (e.g., Cholera)
3. **Food borne** — Through contaminated food (e.g., Typhoid)
4. **Contact** — Direct contact with infected person (e.g., Herpes)
5. **Vector borne** — Through vectors like mosquitoes (e.g., Malaria)

Examples of Infectious Diseases

- * **Tuberculosis** — Bacterium
- * **Malaria** — Protozoan
- * **AIDS** — Virus
- * **Ringworm** — Fungus
- * **Ascariasis** — Helminth

PREVENTION AND CONTROL

- **Personal hygiene** (regular bathing, hand washing)
- **Safe drinking water and proper sanitation**
- **Vaccination and immunization**
- **Balanced diet and regular exercise**
- **Avoiding close contact with infected persons**



Wash Hands



Boil Water



Vaccination



Eat Healthy



INFECTIOUS DISEASES

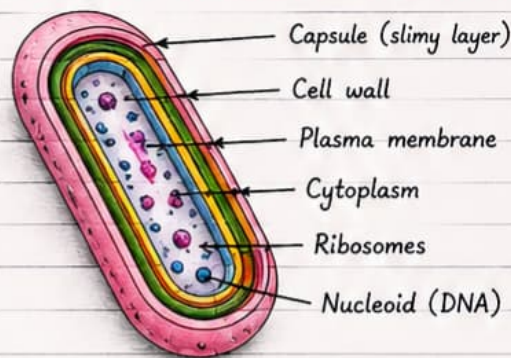
1. BACTERIAL DISEASES

- Bacteria cause a number of diseases in human beings.
- They invade the body, multiply and release toxins which damage body tissues.

Examples and symptoms :

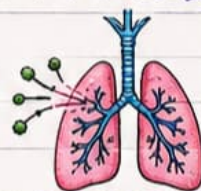
Disease	Causative Bacterium	Symptoms
Tuberculosis (TB)	Mycobacterium tuberculosis	Persistent cough, fever, weight loss, chest pain.
Typhoid	Salmonella typhi	High fever, weakness, stomach pain, loss of appetite.
Cholera	Vibrio cholerae	Severe diarrhoea, vomiting, dehydration.
Pneumonia	Streptococcus pneumoniae	Fever, chills, cough, difficulty in breathing.

Structure of Mycobacterium (TB Bacterium)



Note :

TB is an airborne disease. It spreads when an infected person coughs or sneezes.



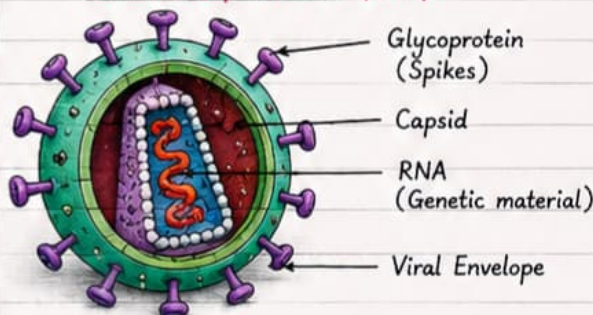
2. VIRAL DISEASES

- Viruses are tiny infectious agents that can replicate **only inside living cells**.

Examples and symptoms :

Disease	Causative Virus	Symptoms
Common Cold	Rhinovirus	Sneezing, runny nose, sore throat.
Influenza (Flu)	Influenza virus	Fever, body ache, cough, headache.
Hepatitis	Hepatitis virus	Jaundice, fatigue, abdominal pain.
Polio	Poliovirus	Fever, sore throat, paralysis (in severe cases).
Measles	Measles virus	High fever, red rashes, cough, running nose.

Structure of a Virus (HIV)



Note :

Viruses do not have their own metabolism. They use the host cell machinery to multiply.



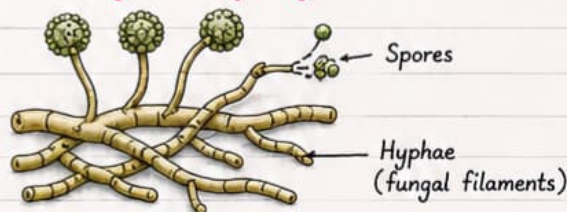
3. FUNGAL DISEASES

- Fungi are eukaryotic organisms. They live on dead and decaying matter as **saprophytes** and cause diseases in humans.

Examples :

- Ringworm (Tinea) – Itching, circular rashes on skin.
- Athlete's foot – Itching, cracks between toes.
- Candidiasis – White patches in mouth, swelling, irritation.

Fungus causing Ringworm



PREVENTION OF INFECTIOUS DISEASES



Wash hands regularly



Drink clean water & eat fresh food



Vaccination and immunization



Use mask and cover mouth



Keep surroundings clean



Proper rest and balanced diet

* A healthy body is the best defence against diseases. *



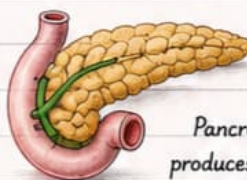
NON-INFECTIOUS DISEASES

- These diseases are **not caused by pathogens**.
- They are usually caused by genetic, nutritional, physiological, or environmental factors.
- Examples :**

Disease	Cause	Symptoms
Diabetes Mellitus	Deficiency of insulin or insulin resistance.	High blood sugar, frequent urination, thirst, fatigue.
Hypertension (High BP)	Stress, high salt diet, obesity, genetic factors.	Headache, dizziness, chest pain.
Asthma	Allergy, pollution, genetic tendency.	Difficulty in breathing, wheezing, coughing.
Cancer	Uncontrolled growth of abnormal cells.	Lump, pain, loss of weight, fatigue.

Some Common Non-Infectious Diseases

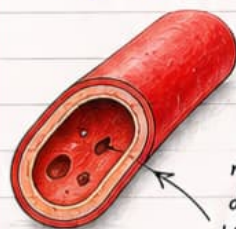
Diabetes Mellitus



Pancreas produces less insulin.

Results in high blood glucose level.

Hypertension



Blood vessel narrows due to high BP.

It increases the risk of heart diseases and stroke.

Asthma



Inflammation and narrowing of airways causes difficulty in breathing.

Cancer



Abnormal cells divide rapidly and may spread to other parts.

3. DEFICIENCY DISEASES

- These diseases are caused by **lack of essential nutrients in the diet**.
- Examples :**

Nutrient Deficiency	Disease	Symptoms
Vitamin A	Night Blindness	Poor vision in dim light, dry eyes.
Vitamin B ₁	Beriberi	Weak muscles, fatigue, nerve problem.
Vitamin C	Scurvy	Bleeding gums, weak immunity.
Vitamin D	Rickets	Weak bones, bone deformities in children.
Iron	Anaemia	Paleness, weakness, tiredness.

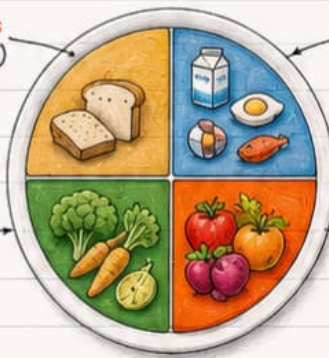
Balanced Diet - Key to Good Health

Carbohydrates
(Energy giving)
e.g., Rice, Roti, Bread

Proteins
(Body building)
e.g., Dal, Milk, Egg, Fish

Vitamins & Minerals
(Protective)
e.g., Fruits, Vegetables

Fats
(Energy storing)
e.g., Nuts, Oils, Ghee



Eat healthy, live healthy 🍏

TREATMENT AND PREVENTION OF DISEASES

1. Treatment

- Depends on the type and cause of disease.
- * **Antibiotics** → Used to kill or stop the growth of bacteria.
- * **Antivirals** → Used against viral infections.
- * **Antifungals** → Used to treat fungal infections.
- * **Symptomatic treatment** → To relieve symptoms and support recovery.

Examples of Medicines

Antibiotic
(Penicillin)



Antiviral
(Acyclovir)



Antifungal
(Clotrimazole)



Pain Reliever
(Paracetamol)



2. Prevention

- ✓ **Personal Hygiene** - Regular bathing, washing hands, clean clothes.
- ✓ **Safe Drinking Water** - Drink clean and boiled water.
- ✓ **Nutritious Food** - Eat balanced diet rich in vitamins and minerals.
- ✓ **Regular Exercise** - Helps in maintaining good health.
- ✓ **Vaccination** - Protects from many infectious diseases.
- ✓ **Avoid Addiction** - Avoid smoking, alcohol and other harmful habits.

★ Remember :

Good health is the **real wealth**.
Healthy today, happy tomorrow! 🍏

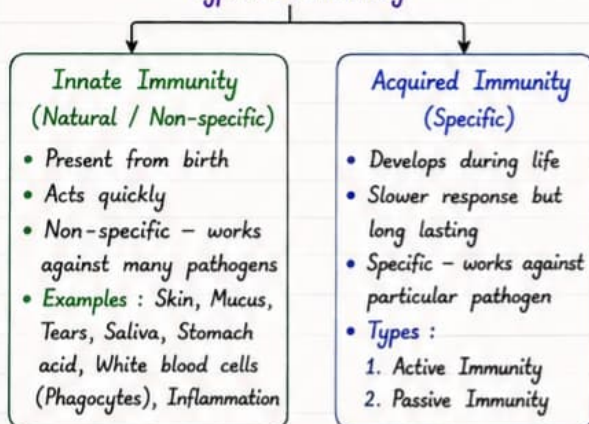


Prevention is better than cure.

4. IMMUNITY

- Immunity is the body's ability to **resist infection and disease**.

Types of Immunity



Acquired Immunity – Detailed

1. Active Immunity

- Body produces its own antibodies.
- Long lasting protection.
- Develops after infection or vaccination.

Examples :

- Chickenpox → lifelong immunity.
- Vaccination (e.g., BCG, Polio vaccine)

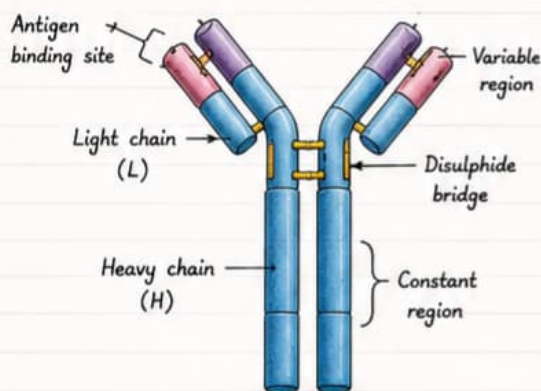
2. Passive Immunity

- Ready-made antibodies are received from outside.
- Short term protection.
- Used in emergencies.

Examples :

- Antibodies from mother to baby via placenta (IgG) or breast milk (IgA)
- Injection of immunoglobulin (e.g., Anti-snake venom)

STRUCTURE OF AN ANTIBODY (IMMUNOGLOBULIN)



- Y-shaped protein produced by B-lymphocytes (Plasma cells).
- Made up of 2 heavy (H) chains and 2 light (L) chains.
- The tip of arms have variable region → binding site for antigen.
- The stem has constant region → determines the type of antibody (IgG, IgM, IgA, IgD, IgE).

Types of Antibodies

Antibody	Where Found	Main Function
IgG	Blood, tissue fluid	Most abundant, provides long term protection
IgM	Blood	First antibody made, strong agglutination
IgA	Mucus, saliva, tears, breast milk	Protects mucous membranes
IgD	On B-cell surface	Acts as receptor on B-cells
IgE	Blood, tissue	Involved in allergies, parasite infections

VACCINATION

- Vaccines contain dead or weakened pathogens.
- They stimulate the immune system to produce antibodies and memory cells.
- Example : BCG (TB), OPV (Polio), Hepatitis B, Measles, DPT.



Vaccination provides active immunity without causing the disease.



IMPORTANT DISEASES AND THEIR CAUSE

Disease	Causative Agent	Mode of Spread	Symptoms	Prevention / Control
AIDS	HIV (Virus)	Sexual contact, blood transfusion, infected needles, mother to child	Fever, weight loss, weakness, frequent infections	Safe sex, no sharing of needles, screening of blood, anti-retroviral drugs
Tuberculosis (TB)	Mycobacterium tuberculosis (Bacterium)	Air borne (coughing, sneezing)	Persistent cough, fever, weight loss, night sweats	BCG vaccine, good ventilation, cough hygiene, complete treatment
Malaria	Plasmodium (Protozoan)	By bite of infected female Anopheles mosquito	Fever with chills, sweating, headache, anemia	Use of mosquito nets, anti-malarial drugs, remove stagnant water
Typhoid	Salmonella typhi (Bacterium)	Contaminated food and water	High fever, stomach pain, weakness, loss of appetite	Clean drinking water, proper cooking, vaccination
Dengue	Dengue virus	By bite of infected Aedes mosquito	High fever, severe headache, body pain, rash, bleeding	Remove stagnant water, use mosquito repellents, wear full sleeves

★ Good Health is the Greatest Wealth.



Quick Remember

Eat Healthy
Stay Clean
Exercise Regularly
Get Vaccinated
Stay Safe, Stay Healthy

5. CANCER

- Cancer is a group of diseases in which **abnormal cells divide uncontrollably** and can spread to other parts of the body.
- It can be caused by genetic changes, radiation, chemicals, viruses and lifestyle factors.

Causes of Cancer

- Genetic mutations (changes in DNA)
- Exposure to radiation (UV rays, X-rays)
- Carcinogenic chemicals (tobacco, smoke, etc.)
- Viral infections (e.g., HPV, Hepatitis B)
- Unhealthy lifestyle (poor diet, lack of exercise, stress, alcohol, smoking)

Types of Cancer (Examples)



Lung Cancer



Breast Cancer



Stomach Cancer



Blood Cancer (Leukemia)

★ Early detection and treatment can increase the chance of recovery.

6. LIFESTYLE DISEASES (NON-COMMUNICABLE DISEASES)

- These diseases are **not** caused by pathogens.
- They are caused by unhealthy lifestyle, stress, poor diet and lack of physical activity.

Disease	Causes	Symptoms
 High Blood Pressure (Hypertension)	Stress, high salt diet, obesity, lack of exercise, smoking.	Headache, dizziness, chest pain, fatigue.
 Diabetes Mellitus	Deficiency of insulin, high sugar diet, obesity.	Frequent urination, increased thirst, fatigue, blurred vision.
 Coronary Heart Disease (CHD)	High fat diet, smoking, high BP, diabetes, stress.	Chest pain, breathing difficulty, heart attack.
 Arthritis	Ageing, obesity, joint injury, genetic factors.	Joint pain, swelling, stiffness, reduced movement.

★ Regular exercise, balanced diet and stress management help in preventing lifestyle diseases.



SUMMARY

- Health is complete physical, mental and social well-being.
- Diseases are of three types - infectious, non-infectious and deficiency diseases.
- Pathogens cause infectious diseases and spread through different modes.
- Prevention is better than cure - adopt clean habits, balanced diet and vaccination.
- Immunity protects the body from diseases.
- Lifestyle diseases can be prevented by healthy habits.
- Early detection and proper treatment lead to a healthy life.

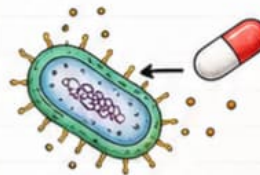


ANTIBIOTICS AND ANTIBIOTIC RESISTANCE

- Antibiotics are medicines produced by microorganisms that **kill or stop** the growth of bacteria.
- Overuse and misuse of antibiotics can lead to **antibiotic resistance**.

How Antibiotics Work

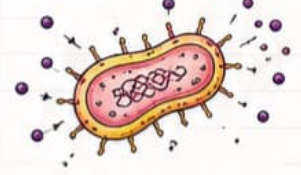
- They target specific structures in bacteria such as cell wall, protein synthesis or DNA.
- They do not work against viruses.



Antibiotic attacking bacteria

Antibiotic Resistance

- Bacteria evolve and develop mechanisms to survive antibiotics.
- This makes the treatment ineffective.



Resistant bacteria survives and multiplies

How to Prevent Antibiotic Resistance

- Use antibiotics only when prescribed by a doctor.
- Complete the full course of medicine.
- Do not use antibiotics for viral infections (like cold, flu).
- Do not share or use leftover antibiotics.



CANCER CELL VS NORMAL CELL

Normal Cell



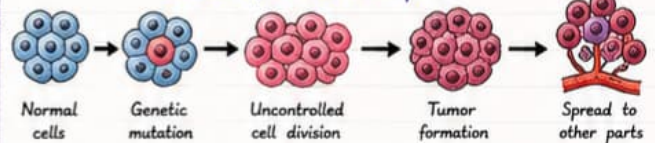
- Normal in size and shape
- Controlled cell division
- Stay in one place
- Do not invade other tissues
- Function normally

Cancer Cell



- Abnormal in size and shape
- Uncontrolled cell division
- Can spread to other parts
- Invade other tissues
- Function abnormally

Process of Cancer Development



★ Cancer can be treated by surgery, radiation therapy, chemotherapy and targeted therapy.



KEY POINTS

- Infectious diseases → caused by pathogens.
- Non-infectious diseases → caused by genetics, nutrition, environment and lifestyle.
- Deficiency diseases → caused by lack of essential nutrients.
- Immunity → natural and acquired, protects from diseases.
- Antibiotics → useful but overuse causes resistance.
- Vaccination → builds immunity and prevents diseases.

